

Machine Learning Models

1. Introduction

Machine Learning (ML) is a field of Artificial Intelligence (AI) that enables systems to learn patterns from data and make predictions or decisions without being explicitly programmed.

An ML model is a mathematical representation of real-world data trained to perform tasks such as prediction, classification, or clustering.

2. Types of Machine Learning

1. Supervised Learning

- Data has input (features) and output (labels).
- Examples:
 - Predicting house prices (Regression).
 - Classifying emails as spam or not spam (Classification).

2. Unsupervised Learning

- Data has only input (no labels).
- Model finds hidden patterns or groups.
- Examples:
 - Customer segmentation.
 - Market basket analysis.

3. Reinforcement Learning

- Agent learns by interacting with an environment.
 - Example:
 - Self-driving cars.
 - Game-playing AI (like AlphaGo).
-

3. Common ML Models

🔑 Linear Regression

- **Type:** Regression
- **Goal:** Predict continuous values using a straight line relationship between features and target.
- **Example:** Predicting house prices.

🔑 Logistic Regression

- **Type:** Classification
- **Goal:** Predict probabilities of classes using a logistic (sigmoid) function.
- **Example:** Predicting if a network traffic is Normal or Attack.

🔑 Decision Tree

- **Type:** Classification/Regression
- **Goal:** Splits data into branches based on feature values.
- **Example:** Fraud detection.

🔑 Random Forest

- **Type:** Ensemble Classification/Regression
- **Goal:** Uses many decision trees and takes majority vote/average.
- **Example:** Intrusion detection (works well on NSL-KDD).

🔑 Support Vector Machine (SVM)

- **Type:** Classification/Regression
- **Goal:** Finds the best separating boundary (hyperplane) between classes.
- **Example:** Classifying emails as spam or not spam.

🔑 Neural Networks

- **Type:** Classification/Regression
- **Goal:** Mimics human brain with layers of “neurons.”
- **Example:** Image recognition, NLP, anomaly detection.

4. Workflow of ML Modeling

1. **Data Collection** – get dataset (e.g., NSL-KDD).
2. **Data Preprocessing** – clean, encode, scale.
3. **Train-Test Split** – divide dataset into training & testing.
4. **Model Selection** – choose ML model (e.g., Logistic Regression).
5. **Training** – fit model on training data.
6. **Evaluation** – test model on unseen data.
7. **Deployment** – use the trained model in real-world applications.

5. Model Evaluation Metrics

- **Regression:** MSE (Mean Squared Error), R^2 Score.

- **Classification:** Accuracy, Precision, Recall, F1-Score, Confusion Matrix.
-

6. Conclusion

ML models provide powerful tools to analyze data, make predictions, and automate decision-making. Choosing the right model depends on the **type of problem (regression/classification/clustering)**, the **quality of data**, and the **required accuracy**.